United States Patent Application Publication	20250276756
Kind Code	A1
Publication Date	September 04, 2025
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ELECTRIC VEHICLE WITH AN ELECTRIC POWERPACK ARRANGEMENT

Abstract

A straddle seat electric vehicle including an electric motor and an electric powerpack including a battery pack including a battery housing including a housing body, a cooling channel extending generally vertically through a center portion of the housing body, two covers selectively connected to the housing body; a plurality of cylindrical battery cells disposed in a two chambers defined by the housing laterally between the cooling channel and the first cover, each battery cell of the first plurality of battery cells extending generally orthogonally to the cooling channel and the first cover; one or more current collectors electrically connected to the first plurality of battery cells, the covers enclosing the current collectors and outer ends of the battery cells being electrically insulated from the at least one first current collector.

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Family ID: **86157482**

Appl. No.: **18/704600**

Filed (or PCT Filed): **October 31, 2022**

PCT No.: **PCT/IB2022/060483**

Related U.S. Application Data

us-provisional-application US 63273435 20211029
us-provisional-application US 63273468 20211029

Publication Classification

Int. Cl.: **B62J43/16** (20200101); **B60L53/16** (20190101); **B60L53/22** (20190101); **B62J43/28** (20200101); **B62M7/04** (20060101); **B62M27/02** (20060101); **H01M10/42** (20060101); **H01M10/613** (20140101); **H01M50/107** (20210101)

U.S. Cl.:

CPC **B62J43/16** (20200201); **B60L53/16** (20190201); **B60L53/22** (20190201); **B62J43/28** (20200201); **B62M7/04** (20130101); **B62M27/02** (20130101); **H01M10/425** (20130101); **H01M10/613** (20150401); **H01M50/107** (20210101); B60L2200/12 (20130101); B60L2210/10 (20130101); B62M2027/025 (20130101); H01M2010/4271 (20130101)

Background/Summary

CROSS-REFERENCE [0001] The present application claims priority from U.S. Provisional Patent Application No. 63/273,435, entitled “Electric Vehicle with a Cooling Arrangement,” filed Oct. 29, 2021, and to U.S. Provisional Patent Application No. 63/273,468, entitled “Electric Vehicle with an Electric Powerpack Arrangement,” filed Oct. 29, 2021, the entirety of both of which is incorporated by reference herein.

FIELD OF TECHNOLOGY
[0002] The present technology relates to vehicles having electric powerpack arrangements, specifically straddle-seat electric vehicles.
BACKGROUND
[0003] Straddle seat vehicles, including motorcycles, all-terrain vehicles, and snowmobiles, are popular transport and recreational vehicles. As the move toward electrification of vehicles progresses, interest in electric versions of straddle seat vehicles increases.
[0004] For the case of straddle seat vehicles, space available for different electric components, including at least a battery assembly, charging components, an electric motor, and components for managing power distribution, is strictly limited. Mounting the components of an electric powerpack in a spatially efficient manner, while still providing convenient access for maintenance, thus remains a challenge.
[0005] There is therefore a desire for powerpack arrangements for electric straddle seat vehicles addressing at least some of the above described disadvantages.
SUMMARY
[0006] It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.
[0007] According to one aspect of the present technology, there is provided an electric vehicle with a powerpack arrangement to arrange battery components in an efficient manner while maintaining access to different components for ease of maintenance. The battery pack is formed from battery modules including a plurality of battery cells connected in series. Each module has a battery management system, and each battery management system is connected to a global battery management system disposed in the battery pack. The battery housing also includes a battery cooling channel defined through a center thereof. The cooling channel is fluidly connected to a liquid cooling system of the vehicle, permitting the battery cells to be cooled during operation while maintaining an efficient spatial arrangement of the battery pack. The electric powerpack further includes an inverter and a charger mounted directly to the battery pack to further minimize the spatial footprint of the powerpack.

[0008] According to one aspect of the present technology, there is provided a straddle seat electric vehicle including a frame; a straddle seat supported by the frame; at least one ground-engaging member operatively connected to the frame; an electric motor being operatively connected to the at least one ground-engaging member for driving the at least one ground-engaging member; an electric powerpack operatively connected to the electric motor, the powerpack including a battery pack including a battery housing including a housing body, a cooling channel defined in the housing body, the cooling channel extending generally vertically through a center portion of the housing body, a first cover selectively connected to the housing body, and a second cover selectively connected to the housing body; a first plurality of cylindrical battery cells disposed in a first chamber defined by the housing laterally between the cooling channel and the first cover, each battery cell of the first plurality of battery cells extending generally orthogonally to the cooling channel and the first cover; at least one first current collector disposed laterally between the first cover and the first plurality of battery cells, the at least one first current collector being electrically connected to the first plurality of battery cells, the first cover enclosing the at least one first current collector and outer ends of the battery cells of the first plurality of battery cells, the first cover being electrically insulated from the at least one first current collector; a second plurality of cylindrical battery cells disposed in a second chamber defined by the housing laterally between the cooling channel and the second cover, each battery cell of the second plurality of battery cells extending generally orthogonally to the cooling channel and the second cover; and at least one second current collector disposed laterally between the second cover and the second plurality of battery cells, the at least one second current collector being electrically connected to the second plurality of battery cells, the second cover enclosing the at least one second current collector and outer ends of the battery cells of the second plurality of battery cells, the second cover being electrically insulated from the at least one second current collector.

[0009] In some embodiments, the battery cells of the first plurality of battery cells are separated into a first plurality of battery modules; the at least one first current collector includes a first plurality of current collectors connected in series; each one of the first plurality of current collectors is electrically connected to a corresponding battery module of the first plurality of battery modules; the battery cells of the second plurality of battery cells is separated into a second plurality of battery modules; the at least one second current collector includes a second plurality of current collectors connected in series; and each one of the second plurality of current collectors is electrically connected to a corresponding battery module of the second plurality of battery modules.

[0010] In some embodiments, the second plurality of battery modules has a greater number of battery modules than the first plurality of battery modules.

[0011] In some embodiments, the first plurality of battery modules includes three first battery modules disposed in the first chamber; the first plurality of current collectors includes three first current collectors; the second plurality of battery modules includes four second battery modules disposed in the second chamber; and the second plurality of current collectors includes four second current collectors.

[0012] In some embodiments, the three first current collectors are electrically connected to one another in series; and the four second current collectors are electrically connected to one another in series.

[0013] In some embodiments, the first plurality of current collectors is a first plurality of printed circuit boards (PCBs); and the second plurality of current collectors is a second plurality of PCBs.

[0014] In some embodiments, each PCB of the first and second pluralities of PCBs includes a module battery management system (BMS) for managing the corresponding battery module of the first and second pluralities of battery modules.

[0015] In some embodiments, the battery pack further includes at least one of a global BMS disposed in the first chamber, the global BMS being electrically connected to the at least one first current collector and the at least one second current collector; and a DC-DC converter disposed in the first chamber, the DC-DC converter being electrically connected to at least one of the at least one first current collector and the at least one second current collector.

[0016] In some embodiments, the electric powerpack further comprises a plurality of electrical components; at least one of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells and disposed on an exterior of the battery housing; and at least one other of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells and disposed on an interior of the battery housing.

[0017] In some embodiments, the electric powerpack further includes at least one of a charger electrically connected to the first and second pluralities of cylindrical battery cells; and an inverter electrically connected to the first and second pluralities of cylindrical battery cells.

[0018] In some embodiments, the electric powerpack comprises both the charger and the inverter.

[0019] In some embodiments, the charger and the inverter are mounted to the battery housing.

[0020] In some embodiments, the housing body includes a first lateral portion, and a second lateral portion connected to the first lateral portion; the first cover being selectively connected to the first lateral portion; the second cover being selectively connected to the second lateral portion; an inner face of the first lateral portion including a first channel form; an inner face of the second lateral portion including a second channel form; the cooling channel is defined by a space created between the inner face of the first lateral portion and the inner face of the second lateral portion; and the center portion of the housing body being formed by an inner part of the first lateral portion and an inner part of the second lateral portion.

[0021] In some embodiments, the battery pack further includes at least one first dielectric layer disposed between inner ends of the first plurality of battery cells and the center portion the housing body; and at least one second dielectric layer disposed between inner ends of the second plurality of battery cells and the center portion the housing body.

[0022] In some embodiments, when in operation, coolant flowing through the cooling channel absorbs heat from inner ends of the first plurality of battery cells and inner ends of the second plurality of battery cells.

[0023] In some embodiments, inner ends of the first plurality of battery cells are in thermal communication with the cooling channel through the housing body; and inner ends of the second plurality of battery cells are in thermal communication with the cooling channel through the housing body.

[0024] In some embodiments, the vehicle further includes a liquid cooling system including a fluid pump; a coolant reservoir; and a cooling circuit formed at least in part by the cooling channel of the battery housing, the cooling channel being fluidly connected to the fluid pump and the coolant reservoir.

[0025] In some embodiments, the at least one ground-engaging member includes a front wheel disposed at least in part forward of the powerpack, and a rear wheel disposed at least in part rearward of the powerpack; and the vehicle is an electric motorcycle.

[0026] In some embodiments, the vehicle is an electric snowmobile and the at least one ground-engaging member includes two skis.

[0027] According to another aspect of the present technology, there is provided a straddle seat electric vehicle including a frame; a straddle seat supported by the frame; at least one ground-engaging member operatively connected to the frame; an electric motor being operatively connected to the at least one ground-engaging member for driving the at least one ground-engaging member; an electric powerpack operatively connected to the electric motor, the powerpack including a battery pack including a battery housing including a housing body, a cooling channel defined in the housing body, the cooling channel extending generally vertically through a center portion of the housing body, a first cover selectively connected to the housing body, and a second cover selectively connected to the housing body; a first plurality of battery modules disposed in a first chamber defined by the housing laterally between the cooling channel and the first cover, each module of the first plurality of battery modules including a plurality of battery cells; a plurality of first current collectors disposed laterally between the first cover and the first plurality of battery modules, each battery module of the first plurality of battery modules being operatively connected to a corresponding one of the first plurality of current collectors, the plurality of battery cells of each battery module of the first plurality of battery modules being connected together in series by the corresponding one of the first plurality of current collectors; a second plurality of battery modules disposed in a second chamber defined by the housing laterally between the cooling channel and the second cover, each module of the second plurality of battery modules including a plurality of battery cells; and a plurality of second current collectors disposed laterally between the second cover and the second plurality of battery modules, each battery module of the second plurality of battery modules being operatively connected to a corresponding one of the second plurality of current collectors, the plurality of battery cells of each battery module of the second plurality of battery modules being connected together in series by the corresponding one of the second plurality of current collectors.

[0028] For the purposes of the present application, terms related to spatial orientation such as forward, rearward, front, rear, upper, lower, left, and right, are as they would normally be understood by a driver of the vehicle sitting therein in a normal driving position with the vehicle being upright and steered in a straight ahead direction.

[0029] Implementations of the present technology each have at least one of the above-mentioned object and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.

[0030] Additional and/or alternative features, aspects and advantages of implementations of the present technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

[0032] FIG. **1** is a top, rear, right side perspective view of an electric motorcycle according to a non-limiting embodiment of the present technology;

[0033] FIG. **2** is a top plan view of the vehicle of FIG. **1**;

[0034] FIG. **3** is left side elevation view of the vehicle of FIG. **1**;

[0035] FIG. **4** is a right side elevation view of the vehicle of FIG. **1**, with body panels having been removed;

[0036] FIG. **5** is a left side elevation view of the vehicle of FIG. **1**, with body panels having been removed;

[0037] FIG. **6** is a top, front, left side perspective view of a frame, a powerpack, a drivetrain, and a swing arm of the vehicle of claim **1**, with a housing cover of the swing arm having been removed;

[0038] FIG. **7** is a top, rear, right side perspective view of the components of FIG. **6**;

[0039] FIG. **8** is a top plan view of the components of FIG. **6** with the housing cover of the swing arm;

[0040] FIG. **9** is a top, front, left side perspective view of the powerpack, the motor, and the cooling circuit components of the vehicle of FIG. **1**;

[0041] FIG. **10** is a perspective, exploded view of portions of the powerpack of FIG. **6**, with battery

cells having been removed;
[0042] FIG. **11** is a cross-sectional view of portions of the powerpack of FIG. **6**, taken along line **11-11** of FIG. **8**;
[0043] FIG. **12** is a rear, left side perspective view of a battery pack of the powerpack of FIG. **6**;
[0044] FIG. **13** is a top, rear, right perspective, partially-exploded view of the battery pack of FIG. **12**;
[0045] FIG. **14** is a top, rear, right side perspective view of an inverter and a rigid connector of the powerpack of FIG. **6**;
[0046] FIG. **15** is a top, rear, left side perspective view of the battery pack of FIG. **12**, with a battery pack housing having been removed;
[0047] FIG. **16** is a left side elevation view of the housing-removed battery pack of FIG. **15**;
[0048] FIG. **17** is a right side elevation view of the housing-removed battery pack of FIG. **15**;
[0049] FIG. **18** is a perspective view of a battery module of the battery pack of FIG. **15**;
[0050] FIG. **19** is a partial, close-up view of the battery module of FIG. **18**;
[0051] FIG. **20** is a perspective, exploded view of the battery module of FIG. **18**; and
[0052] FIG. **21** is a right side elevation view of an electric snowmobile according to another non-limiting embodiment of the present technology.
[0053] It should be noted that, unless otherwise explicitly specified herein, the drawings are not necessarily to scale.

DETAILED DESCRIPTION

[0054] The present technology will be described herein with respect to a straddle-seat electric vehicles, specifically a two-wheeled electric motorcycle **100** and an electric snowmobile **300**. Aspects of the present technology could also be implemented in different straddle-seat electric vehicles, such as three-wheeled electric vehicles and all-terrain vehicles (ATVs).
[0055] While the motorcycle **100** illustrated herein is a trail style electric motorcycle **100**, it is contemplated that motorcycles according to the present technology could vary by a plurality of vehicle characteristics. These vehicle characteristics could include, but are not limited to, a rider posture configuration (also referred to as a rider position), a motorcycle type, tire type, a wheelbase, a steering arrangement, a weight distribution, a squat ratio, a rake angle, a seat height, and a mechanical trail. The rider posture configuration, or rider position, is the relative spacing and position of a rider's hands (when holding the handlebars), the rider's feet (when positioned on the footrests) and the rider's buttocks (when the rider is seated on a seat of the motorcycle). The steering arrangement could also vary and can be described by a variety of parameters, including but not limited to: a length of front suspension travel, a length of rear suspension travel, a front suspension stiffness, a rear suspension stiffness, a front and/or rear wheel size, rake angle, mechanical trail, triple clamp offset, squat ratio, and wheel base.
[0056] With reference to FIGS. **1** to **5**, the electric motorcycle **100**, referred to herein as the vehicle **100**, has a front end **102**, a rear end **104**, and a longitudinal centerplane **103** defined consistently with the forward travel direction of the vehicle **100**.
[0057] The vehicle **100** has a frame **110**, shown in additional detail in FIGS. **6** to **8**. The frame **110** includes a front suspension receiving portion **112**, specifically a tube **112**, sometimes referred to as a “head tube”, for receiving therethrough a front fork assembly **124** (described in more detail below). Extending rearward from the tube **112**, the frame **110** includes forward frame members **114**. In the illustrated embodiment, there are six forward frame members **114** (three on each side of the centerplane **103**) but it is contemplated that the specific number and arrangement of forward frame members **114** could vary.
[0058] The frame **110** also includes two upper intermediate frame members **116** extending rearward from the forward members **114**. The frame members **116** are generally hockey-stick shaped, with rear portions of the members **116** curving rearward and downward from generally horizontal forward portions of the members **116**. In different embodiments, the frame members **116** could be

differently shaped.

[0059] The frame **110** further includes two lower intermediate frame members **118** connected to rear ends of the frame members **116**. The frame members **118** extend generally vertically along left and right sides of the vehicle **100**. The frame members **118** are generally in the shape of flattened boomerangs, but the particular shape could vary.

[0060] The frame **110** further includes a rear frame structure **120** (FIGS. **4** and **5**) connected to and extending rearward and slightly upward from the upper intermediate frame members **116** and the lower intermediate frame members **118** (omitted from FIGS. **6** and **7**). The rear frame structure **120** is also referred to as a seat support structure **120**. It is contemplated that the frame **110** could include additional frame members, including but not limited to additional intermediate frame members and additional rear frame members.

[0061] The vehicle **100** is a two-wheeled vehicle **100** including a front wheel **121** and a rear wheel **127**. The front wheel **121** and the rear wheel **127** each have a tire secured thereto. The front wheel **121** and the rear wheel **127** are centered with respect to the longitudinal centerplane **103**.

[0062] The front wheel **121** is connected to the frame **110** by a front suspension assembly **123**. The front suspension assembly **123** includes a front fork assembly **124** for supporting the front end **102** of the vehicle **100**. The front fork assembly **124** includes a triple clamp assembly **125** connected to the tube **112** of the frame **110**. The front fork assembly **124** includes a pair of front shocks **122** connected to the triple clamp assembly **125**. The front wheel **121** of the front fork assembly **124** is connected to a bottom portion of the pair of front shocks **122**.

[0063] The rear wheel **127** mounted to the frame **110** by a rear suspension assembly **130**. The rear suspension assembly **130** includes a swing arm **132** and a shock absorber **136**. The swing arm **132** is pivotally mounted at a front thereof to the frame **110**, specifically to lower portions of the lower intermediate frame portions **118**. As best seen in FIG. **6**, the swing arm **132** includes a swing arm housing **134**, in which is disposed an electric motor **160** and a drivetrain **170**. The swing arm **132** includes a housing cover **137** selectively removable from the housing **134**. The swing arm housing **132** and the housing cover **137** enclose the drivetrain **170**. When the housing cover **137** is in place, the drivetrain **170** is bathed in lubricant within the swing arm **132**.

[0064] The rear wheel **127** is rotatably mounted to the rear end of the swing arm **132** which extends on a left side of the rear wheel **127**. The shock absorber **136** is connected between the swing arm **132** and the frame **110**, specifically to the intermediate frame members **116**. It is contemplated that the relative arrangement of the shock absorber **136** and the frame **110** could vary in different embodiments. The electric motor **160** and the drivetrain **170** will be described in more detail below.

[0065] The vehicle **100** has a straddle seat **140** mounted to the frame **110**, specifically to the rear frame structure **120**, and disposed along the longitudinal centerplane **103**. In the illustrated implementation, the straddle seat **140** is intended to accommodate a single adult-sized rider, i.e. the driver. It is however contemplated that the seat **140** could be longer or that a passenger seat portion could be connected to the rear frame structure **120** in order to accommodate a passenger behind the driver. Depending on the particular implementation, it is also contemplated that the seat **140** could be supported by an assembly of frame members or tubes, a molded portion integrally connected to the seat **140**, or body panels of the motorcycle **100**.

[0066] The vehicle **100** further includes a plurality of body panels **142** for forming the body of the vehicle **100**, illustrated in FIGS. **1** to **3**. The body panels **142** are connected to and supported by the frame members **114**, **116**. The body panels **142** enclose and protect some internal components of the vehicle **100** such as a powerpack **200** (described further below). The vehicle **100** also includes a front fender **144** disposed at the front of the vehicle **100** and extending partially over the front wheel **121**. Rearward of the seat **140**, the vehicle **100** also has rear fender panels **146** extending at least partially over rear wheel **127**. The vehicle **100** includes front headlights **145** attached to the front fork assembly **124** and electrically connected to a battery pack **210** (described further below). The vehicle **100** also has rear braking and indicator lights **147** supported by the rear panels **146** and electrically connected to the battery pack **210**.

[0067] Depending on the particular embodiment, especially between different motorcycle types (trail-style motorcycle or cruiser-style motorcycle for example), the body panels **142** and the fenders **144**, **146** could be different in shape and number. For example, some embodiments of the vehicle **100** could include a mud flab connected to a rear edge of one of the body panels **142**. It is further contemplated that one or both of the fenders **144** and rear panels **146** could be omitted in some cases.

[0068] A driver footrest **126** is disposed on either side of the vehicle **100** and vertically lower than the straddle seat **140** to support the driver's feet. The driver footrests **126** are connected to the frame members **118**. It is contemplated that the footrests **126** could be implemented in various forms other than those illustrated, including but not limited to pegs and footboards. It is contemplated that the vehicle **100** could also be provided with one or more passenger footrests disposed rearward of the driver footrest **126** on each side of the vehicle **100**, for supporting a passenger's feet when a passenger seat portion for accommodating a passenger is connected to the vehicle **100**. A brake pedal **128** is connected to the right driver footrest **126** for braking the vehicle **100**. The brake pedal **128** extends upwardly and forwardly from the right driver footrest **126** such that the driver can actuate the brake pedal **128** with a front portion of the right foot while a rear portion of the right foot remains on the right driver footrest **126**.

[0069] With reference to FIGS. **3** to **5**, each of the front wheel **121** and the rear wheel **127** is provided with a brake assembly **90**. The brake assemblies **90** of the wheels **121**, **127**, along with the brake pedal **128**, form a brake system **92**. Each brake assembly **90** is a disc-type brake mounted onto the spindle of the respective wheel **121** or **127**. Other types of brakes are contemplated. Each brake assembly **90** includes a rotor **94** mounted onto the wheel hub and a stationary caliper **96** straddling the rotor **94**. The brake pads (not shown) are mounted to the caliper **96** so as to be disposed between the rotor **94** and the caliper **96** on either side of the rotor **94**. The brake pedal **128**, as well as a hand brake **155** described below, are operatively connected to the brake assemblies **90** provided on each of the front wheel **121** and the rear wheel **127**. The brake system **92** further includes a regenerative braking system (not shown) that uses the electric motor **160** as a generator to charge battery cells of the battery pack **210** while slowing the vehicle **100**.

[0070] Returning to FIGS. **1** to **5**, the vehicle **100** includes a handlebar assembly **152** operatively connected to the front fork assembly **124** and disposed in front of the seat **140**. The handlebar assembly **152** is used by the rider to turn the front wheel **121**, via the front fork assembly **124**, to steer the vehicle **100**. Specifically, the handlebar assembly **152** is connected to a top end of the triple clamp assembly **125**. The handlebar assembly **152** and the triple clamp assembly **125** define a steering axis about which the front wheel **121** turns to steer the vehicle **100**. A twist-grip throttle **153** is operatively connected on the right side of the handlebar assembly **152** for controlling vehicle speed. It is contemplated that the twist-grip throttle **153** could be replaced by a throttle lever or some other type of throttle input device. The twist-grip throttle **153** could be disposed on the left side of the handlebar assembly **152** in some embodiments. The handlebar assembly **152** also includes a hand brake **155** on a right side for activating the brake assemblies **90**.

[0071] It is contemplated that the vehicle **100** could include a variety of different features excluded from discussion here, including but not limited to: a windscreen, radio and/or navigational systems, and luggage rack systems.

[0072] The vehicle **100** further includes an electronic powerpack **200**, an electric motor **160**, and a drivetrain **170** for driving the vehicle **100**, specifically the rear wheel **127**. With additional reference to FIGS. **6** to **8**, the electric motor **160** is disposed in the swing arm **132**. As the swing arm **132** pivots relative to the frame **110**, the motor **160** moves with the swing arm **132**. In the present embodiment, the motor **160** is a three-phase electric motor **160**. It is contemplated that different types of motors could be used in some embodiments.

[0073] The motor **160** is operatively connected to the drivetrain **170** disposed in the swing arm **132**, as is illustrated in FIG. **6**, in which the housing cover **137** has been removed to show the drivetrain **170**. An output shaft **161** of the motor **160** has a gear **171** disposed thereon for engaging the drivetrain **170**. The drivetrain **170** includes a gear wheel **172**, a front sprocket **174** connected to the gear wheel **172** and a rear sprocket **176**. The sprocket **174** engages a belt **178** which in turn engages the sprocket **176**. A belt tensioner **177** presses against the lower part of the belt **178** between the sprockets **174**, **176**.

[0074] Power is provided to the motor **160** by the electronic powerpack **200**. Illustrated in additional detail in FIGS. **6** to **8**, the powerpack **200** is supported by the frame **110**. In the present embodiment, the powerpack **200** is connected to the frame members **118** but different embodiments could have different structural arrangements for connecting the powerpack **200** to the frame **110**.

[0075] With additional reference to FIGS. **9** to **13**, the powerpack **200** includes a battery pack **210**. The battery pack **210** includes a battery housing **220**. The battery housing **220** is fastened to the frame members **118** to support the powerpack **200** in the illustrated embodiment. The battery pack **210** includes a plurality of battery cells **230** housed in the battery housing **220**. The battery pack **210** is described in more detail below.

[0076] The powerpack **200** includes a charger **250** connected to the battery pack **210**. The charger **250** includes a charger housing **252** surrounding internal electronic components (not shown) of the charger **250**. The charger **250** is mounted to the battery housing **220**. Specifically, the charger housing **252** is fastened to the battery housing **220** and is disposed on a top side of the battery housing **220**. It is contemplated that the location of the charger **250** relative to the battery pack **210** could vary.

[0077] The charger **250** is electrically connected to the battery cells **230** for supplying charge to the battery cells **230**; the connection arrangement is described in more detail below. The vehicle **100** includes a socket **258** electrically connected to the charger **250** for electrically connecting to an external power source for providing electricity to the charger **250** for charging the battery cells **230**. The socket **258** is disposed generally rearward of the charger **250** and extends at least partially through one of the body panels **142**, but the specific location could vary.

[0078] The powerpack **200** also includes an inverter **260** disposed on a left side of the battery pack **210**. The inverter **260** includes an inverter housing **262** which is fastened to the battery housing **220**, specifically along a left side of the battery housing **220**. As such, the inverter **260** is mounted to the battery housing **220**. In some embodiments, it is contemplated that the inverter **260** could be disposed on a different side of the battery pack **210**.

[0079] In order to electrically connect to the battery cells **230** in the battery pack **210**, the inverter **260** includes an electric connector **261** disposed on an exterior of the invertor housing **262** (see FIG. **10**). The battery pack **210** includes an electric connector **215** electrically connected to the battery cells **230** and disposed on an exterior of the battery housing **220**, specifically on a left side of the housing **220** (see FIG. **17**). When the vehicle **100** is in operation, the inverter **260** receives electric power from the battery cells **230** via the electric connector **215** and the electric connector **261**.

[0080] The connector **215** is arranged to receive the connector **261** of the inverter **260**, such that the electric connector **215** and the electric connector **261** are selectively connected together for managing electricity flow from the battery pack **210** to other electronic components of the vehicle **100**. As can be seen in at least FIG. **6**, the inverter **260** is electrically connected to the three-phase motor **160** via three power cords **165** connected to three outlets **269** of the inverter **260**. The number of cords or type of electrical connection between the inverter **260** and the motor **160** could vary in different embodiments. While the inverter **260** connects directly to the battery pack **210** in the present embodiment, it is contemplated that the inverter **260** could be separated and spaced from the battery pack **210** and electrically connected to the battery cells **230** via power cords or the like.

[0081] In the illustrated embodiment, the vehicle **100** further includes cooling system including a liquid-based cooling circuit **290** (shown schematically in FIG. **5**) for managing the temperature of the powerpack **200** during operation of the vehicle **100**. Heat transfer in the cooling circuit **290** is provided by a liquid coolant, generally glycol-water coolant, but it is contemplated that other coolants could be used. It is noted that while liquid coolant is provided, some gases may also be present in the cooling circuit **290**, due to phase transitions or air infiltrations. In the present embodiment, the cooling circuit **290** cools the powerpack **200** and the motor **160**, but the cooling circuit **290** could be limited to the powerpack **200** in some embodiments. It is also contemplated that some components of the powerpack **200** could be omitted from the cooling circuit **290** and cooling could be provided by other means. For example, some components of the vehicle **100** could be cooled through air cooling.

[0082] The cooling circuit **290** is formed, in part, by a charger cooling channel **254** defined the charger housing **252** (see FIG. **11**) and an inverter cooling channel **264** formed by the inverter housing **262** (see FIG. **14**).

[0083] With reference to FIGS. **7** to **9**, the cooling system of the vehicle **100** also includes a coolant pump **278** for circulating cooling through and forming a portion of the coolant circuit **290**. In the illustrated embodiment, the coolant pump **278** is electrically connected to the battery pack **210** for powering the pump **278** (described further below). The pump **278** is disposed partially rearward of the powerpack **200** in the present embodiment, although it is contemplated that placement of the pump **278** could vary.

[0084] The vehicle **100** further includes two radiators **280**, **282** for cooling the coolant fluid. In the illustrated embodiment, the radiators **280**, **282** are disposed partially forward of the powerpack **200**, although exact placement could vary in different embodiments. The vehicle **100** includes two flexible tubes **285**, with each tube **285** being connected between a corresponding one of the radiators **280**, **282** and the pump **278**, forming a portion of the cooling circuit **290**. In the illustrated embodiment, the right radiator **282** includes a fan **284** connected to the housing of the radiator **282** to aid in increasing the cooling efficiency of the radiator **282**. Depending on the embodiment, it is contemplated that the left radiator **280** could additionally or alternatively include a fan connected thereto. It is also contemplated that the fan **284** could be omitted in some cases.

[0085] The cooling system also includes a coolant reservoir **270** connected to the powerpack **200** and fluidly connected to the cooing circuit **290**. The reservoir **270** receives liquid coolant therein and supplies coolant to the cooling circuit **290**. The reservoir **270** includes a reservoir cap **272** selectively connected thereto. The reservoir **270** provides for coolant to be refilled or supplemented if necessary. When the cap **272** is removed, additional coolant fluid can be added to the reservoir **270** to supplement the fluid level of coolant in the cooling circuit **290**. It is contemplated that the reservoir **270** could be omitted in some embodiments. The reservoir **270** is disposed partially forward of the powerpack **200**, but it is contemplated that the exact positioning of the reservoir **270** could vary in different embodiments. The radiators **280**, **282** are connected to the coolant reservoir **270** via two flexible hoses **287**.

[0086] Returning to the battery pack **210** and with additional reference to FIGS. **10** to **13**, the battery housing **220** includes a housing body **227**, forming a center portion of the housing **220**. As is illustrated in FIG. **13**, the housing body **227** includes a left lateral portion **227**A and a right lateral portion **227**B connected together to form the body **227**. In the illustrated embodiment the left and right lateral portions **227**A, **227**B are selectively connected together via threaded fasteners (not shown). It is contemplated that the left and right lateral portions **227**A, **227**B could be otherwise connected together in different manners. In the present embodiment, the housing body **227** is formed from aluminum, but could be formed from different materials, including but not limited to plastic or other metals.

[0087] The battery housing **220** defines a battery cooling channel **226** therein, specifically through a center portion of the housing body **227**. As can be seen in FIGS. **11** and **13**, the battery cooling channel **226** includes a plurality of fins extending inward from the housing **220** and coolant fluid flows along a longitudinal direction through a center portion **227** of the housing **220**, along the direction of the centerplane **103**, as well as along a vertical/lateral plane of the vehicle **100** (orthogonal to the centerplane **103**), descending toward a battery cooling channel outlet **217** (FIG. **15**). An inner face of the left lateral portion **227**A includes a first channel form **226**A (FIG. **13**) formed thereon and an inner face of the right lateral portion **227**B includes a second channel form **226**B (FIG. **11**) formed thereon. The cooling channel **226** is defined by the space created between the channel forms **226**A, **226**B of the inner faces of the left and right lateral portions **227**A, **227**B. The battery cooling channel **226** is fluidly connected, through the outlet **217** as well as a channel inlet **219** (FIG. **15**) to the coolant reservoir **270**, the coolant pump **278**, the charger cooling channel **254**, and the inverter cooling channel **264**.

[0088] The battery housing **220** also includes a left side cover **221** selectively connected to the housing body **227**, specifically selectively connected to the left lateral portion **227**A. The housing **220** similarly includes a right side cover **223** selectively connected to the housing body **227**, specifically selectively connected to the right lateral portion **227**B. Each cover **221**, **223** is selectively fastened to the housing body **227** to encase the components therein. It is contemplated that the covers **221**, **223** could be selectively connected to the housing body **227** in different manners, including for example by tabs. A left chamber **225** is formed between the center portion of the housing body **227** and the left cover **221**. A right chamber **229** is formed between the center portion of the housing body **227** and the right cover **223**. The left and right chambers **225**, **229** are shown in the exploded view of FIG. **10**.

[0089] The battery pack **210** includes the plurality of battery cells **230**, as mentioned above. As can be seen in FIGS. **15** to **17**, the battery cells **230** are separated (grouped) into battery modules **235**. In the illustrated embodiment, the battery pack **210** includes seven (7) modules **235**; the exact number of modules could vary in different embodiments. The arrangement of the battery cells **230** into modules **235** is described further below.

[0090] The battery modules **235** are separated into two banks of modules: a left bank **233** having three modules **235** disposed in the left chamber **225**, and a right bank **234** having four modules **235** disposed in the right chamber **229**. Depending on the embodiment, the left and right banks **233**, **234** of modules **235** could include more or fewer modules **235**. It is also contemplated that the left and right banks **233**, **234** could have equal numbers of modules **235**. The left bank **233** has more modules **235** than the right bank **234** in the present embodiment, but it is contemplated that the right bank **234** could have more modules **235** than the left bank **233**. Each bank **233**, **234** of modules **235** is electrically connected together in series by a plurality of bus bars **237**.

[0091] With additional reference to FIGS. **17** to **19**, each module **235**, also referred to as battery bricks **235**, includes a portion of the battery cells **230**. The battery cells **230** are cylindrical battery cells **230**, specifically **490** (four hundred ninety) total cylindrical battery cells **230**. In the present embodiment, the battery cells **230** are 3.5V cylindrical cells, such as LG™ M50L cells, but it is contemplated that different versions of cells could be used in some embodiments. For example, battery cells could vary in nominal energy capacity, usable energy capacity, discharge rate, cell chemistry and cell type.

[0092] A long axis of each cylindrical battery cell **230** extends generally laterally in the battery pack **210**, generally orthogonal to the cooling channel **226**, the centerplane **103**, and the lateral outer surfaces of the covers **221**, **223**. In the illustrated embodiment, each module **235** includes seventy cylindrical battery cells **230** arranged in five parallel rows of fourteen cells **230**. It is contemplated that the battery cells **230** could be grouped into larger or smaller modules **235**, depending on the embodiment.

[0093] Each module **235** includes a support matrix **247** for supporting the battery cells **230**. The support matrix **247** is formed from a rigid, electrically isolating material, rigid plastic in the present embodiment. The matrix **247** defines therein seventy generally cylindrical cavities **249** for receiving the battery cells **230** in the support matrix **247**. Inner ends of the cells **230** are affixed to and sealed in the support matrix **247** by epoxy **246**. Outer ends of the battery cells **230** are restrained by an outer end **247**A of the matrix **247**, although it is contemplated that other structures could be utilized to maintain the battery cells **230** in the support matrix **247** in different embodiments.

[0094] Each module **235** includes a current collector **245** connected to the support matrix **247**. Each current collector **245** is arranged to collect current from the battery cells **230** of each module **235** when the battery pack **210** is in powering mode and inversely for distributing charge to each battery cell **230** in a given module **235** when the battery pack **210** is in charging mode. In the current embodiment, the current collector **245** of each module is a printed circuit board (PCB) **245** sized and arranged to cover an external side of the corresponding module **235**. In some embodiments, it is contemplated that one current collector electrically connected to multiple modules **235** could be used. In embodiments having a single block of battery cells **230** in each bank **233**, **234**, for instance, only one current collector/PCB **245** could be used.

[0095] As is mentioned above, the modules **235** are electrically connected together in series. Specifically, the three PCBs **245** of the left bank **233** are electrically connected to one another in series via the bus bars **237** and the four PCBs **245** of the right bank **234** are electrically connected to one another in series via the bus bars **237**.

[0096] In order to impede electric discharge from the battery modules **235** onto the battery housing **220**, the left and right covers **221**, **223** enclosing the PCBs **245** and outer ends of the battery cells **230** are electrically insulated from the PCBs **245**. In the present embodiment, the left and right covers **221**, **223** are spaced from the PCBs **245**. In different embodiments, it is contemplated that additional materials or components could be implemented to electrically insulate the left and right covers **221**, **223** from the battery modules **235**.

[0097] In the illustrated embodiment of the PCBs **245**, each PCB **245** defines a plurality of apertures therein to allow connection to the battery cells **230** to the PCB **245**. Each PCB **245** includes a plurality of wire bonds **244** for connecting the battery cells **230** to the PCB **245**, and in turn connecting the battery cells **230** together in series. Each wire bond **244** connects an outer perimeter of one battery cell **230** (the negative terminal) to an inner portion of a neighboring battery cells **230** (the positive terminal). The apertures in the PCB **245** are shaped to allow connection to the negative terminal of each battery cell **230** at a specific location, and otherwise covers remaining portions of the negative terminal. The apertures in the PCB **245** generally leave the positive terminals of each battery cell **230** exposed, but it is contemplated that portions of the positive terminal could be covered by the PCB **245** as well.

[0098] Each PCB **245** also includes a module battery management system (module BMS) **243** for managing and monitoring operations of the corresponding battery module **235**. While each module **235** includes a corresponding module BMS **243** in the present embodiment, it is contemplated that some PCBs **245** could omit the corresponding module BMS **243**.

[0099] Each module **235** further includes a dielectric layer **248** disposed inward of the inner ends of the battery cells **230** and bonded to the cells **230** and the matrix **247** by the epoxy layer **246**. As can be seen in FIGS. **18** and **20**, the dielectric layer **248** forms an inner most surface (as the module **235** is mounted in the battery pack **210**) of the module **235**. When each module **235** is installed in the battery pack **210**, the dielectric layer **248** forms an insulating layer to electrically isolate the battery cells **230** from the housing body **227**. In the illustrated embodiment, the dielectric layer **248** is formed by a sheet of electrical insulation paper, although different materials could be used.

[0100] It is contemplated that in some embodiments, the battery pack **210** could include one dielectric layer disposed between inner ends of the battery cells **230** of the left bank **233** and the housing body **227** and another dielectric layer disposed between inner ends of the battery cells **230** of the right bank **234** and the housing body **227**. In such an embodiment, for instance, the dielectric layers could be applied to a majority or an entirety of inner faces of the housing body **227**.

[0101] By being disposed in the center portion **227** of the housing **220**, the channel **226** is in thermal communication with banks of battery cells **230** disposed on both a right side of the channel **226** and a left side of the channel **226**. Specifically, inner ends of the battery cells **230** of each bank **233**, **234** are in thermal communication with the cooling channel **226** through the housing body **227**. When in operation, coolant flowing through the battery cooling channel **226** absorbs heat from inner ends of the battery cells **230** on each lateral side of the battery cooling channel **226**. The dielectric layers **248** of the modules **235** electrically insulate the housing body **227**, while permitting heating to flow from the battery cells **230** to the coolant in the cooling channel **226**.

[0102] Returning to FIGS. **15** to **17**, the battery pack **210** also includes a global battery management system (global BMS) **240** disposed in the left chamber **225**, connected to the housing body **227**. The global BMS **240** is electrically connected to the PCBs **245** and the module BMS **243** of each module **235** via the bus bars **237**. A plurality of communication wires **238** are also included to communicatively connect the global BMS **240** to each of the PCBs **245** and the corresponding module BMS **243**. It is contemplated that the individual BMSs included in each PCB **245** could be omitted in some embodiments, with management of each module **235** being managed by the global BMS **240**.

[0103] The battery pack **210** further includes a DC-DC converter **242** disposed in the left chamber **225**. The DC-DC converter **242** is electrically connected to battery modules **235**, via the global BMS **240**, in order to receive electricity therefrom. The DC-DC converter **242** is arranged to provide power from the battery cells **230** to different electrical components of the vehicle **100** (other than the motor **160**) that operate at a lower voltage. In the present embodiment, the DC-DC converter **242** provides power to the front and rear lights **145**, **147** and the pump **278**, as well as on-board control units (not shown). It is contemplated that the DC-DC converter **242** could manage power flow from the battery cells **230** to other electronic components of the vehicle **100**, including but not limited to: a dashboard display and a sound system.

[0104] For ease of assembly and efficiency of space utilization, the global BMS **240** and the DC-DC converter **242** are disposed in an interior of the battery housing **220**, although the placement could vary in different embodiments. As can be seen from at least FIGS. **15** and **16**, the global BMS **240** and the DC-DC converter **242** are disposed in the battery pack **210** above the modules **235** of the left bank **233**. The global BMS **240** and the DC-DC converter **242** occupy a volume within the housing **220** of approximately the same footprint as a battery module **235**. The housing **220** is generally sized and shaped for receiving eight battery modules **235**; in the present embodiment, the global BMS **240** and the DC-DC converter **242**, in combination with the inverter **260** mounted outside the housing **220** but within generally the same footprint, thus occupy the space within the housing **220** of an “eighth module” which is omitted.

[0105] With reference to FIG. **21**, another non-limiting embodiment of a vehicle utilizes the powerpack **200** described above, specifically an electric snowmobile **300**. The snowmobile **300** includes a forward end **312** and a rearward end **314** that are defined consistently with a forward travel direction of the snowmobile **300**. The snowmobile **300** includes a frame **316** that includes a tunnel **318**, a motor cradle portion **320** and a front suspension assembly portion **322**.

[0106] The snowmobile **300** includes the powerpack **200** (shown schematically) and an electric motor **324** (shown schematically), carried by the motor cradle portion **320** of the frame **316**. Two skis **326** (a right ski **326** being illustrated) are positioned at the forward end **312** of the snowmobile **300** and are attached to the front suspension assembly portion **322** of the frame **316** through front suspension assemblies **328**. A steering device in the form of handlebar **336** is attached to the upper end of the steering column **334** to allow a driver to rotate the skis **326**, in order to steer the snowmobile **300**.

[0107] An endless drive track **338** is disposed generally under the tunnel **318** and is operatively connected to the motor **324**. The endless drive track **338** is driven to run about a rear suspension assembly **342** for propulsion of the snowmobile **300**. The rear suspension assembly **342** includes a pair of slide rails **344** in sliding contact with the endless drive track **338**. The rear suspension assembly **342** also includes a plurality of shock absorbers **346**. Suspension arms **348** and **350** are provided to attach the slide rails **344** to the frame **316**. A plurality of idler wheels **352** are also provided in the rear suspension assembly **342**. Other types and geometries of rear suspension assemblies are also contemplated.

[0108] At the forward end **312** of the snowmobile **300**, fairings **354** enclose the powerpack **200**. The fairings **354** include a hood and one or more side panels that can be opened to allow access to the powerpack **200** and/or the motor **324** when this is required, for example, for inspection or maintenance of the powerpack **200** and/or the motor **324**. A windshield **356** is connected to the fairings **354** near the forward end **312** of the snowmobile **300**. Alternatively, the windshield **356** could be connected directly to the handlebar **336**. The windshield **356** acts as a wind screen to lessen the force of the air on the driver while the snowmobile **300** is moving forward.

[0109] A straddle seat **358** is positioned over the tunnel **318**. Two footrests **360** are positioned on opposite sides of the snowmobile **300** below the seat **358** to accommodate the driver's feet. A snow flap **319** is disposed at the rear end **314** of the snowmobile **300**. The tunnel **318** consists of one or more pieces of sheet metal arranged to form an inverted U-shape that is connected at the front to the motor cradle portion **320** and extends rearward therefrom.

[0110] The snowmobile **300** has other features and components which would be readily recognized by one of ordinary skill in the art, further explanation and description of these components will not be provided herein.

[0111] The vehicles **100**, **300** implemented in accordance with some non-limiting implementations of the present technology can be represented as follows, presented in numbered clauses.

[0112] CLAUSE 1. A straddle seat electric vehicle (**100**, **300**) comprising: [0113] a frame (**110**); [0114] a straddle seat (**140**) supported by the frame (**110**); [0115] at least one ground-engaging member (**121**, **127**) operatively connected to the frame (**110**); [0116] an electric motor (**160**) being operatively connected to the at least one ground-engaging member (**121**, **127**) for driving the at least one ground-engaging member (**121**, **127**); [0117] an electric powerpack (**200**) operatively connected to the electric motor (**160**), the powerpack (**200**) comprising: [0118] a battery pack (**210**) comprising: [0119] a battery housing (**220**) including: [0120] a housing body (**227**), [0121] a cooling channel (**226**) defined in the housing body (**227**), the cooling channel (**226**) extending generally vertically through a center portion of the housing body (**227**), [0122] a first cover (**221**) selectively connected to the housing body (**227**), and [0123] a second cover (**223**) selectively connected to the housing body (**227**); [0124] a first plurality of cylindrical battery cells (**230**) disposed in a first chamber (**225**) defined by the housing (**220**) laterally between the cooling channel (**226**) and the first cover (**221**), each battery cell (**230**) of the first plurality of battery cells (**230**) extending generally orthogonally to the cooling channel (**226**) and the first cover (**221**); [0125] at least one first current collector (**245**) disposed laterally between the first cover (**221**) and the first plurality of battery cells (**230**), the at least one first current collector (**245**) being electrically connected to the first plurality of battery cells (**230**), [0126] the first cover (**221**) enclosing the at least one first current collector (**245**) and outer ends of the battery cells (**230**) of the first plurality of battery cells (**230**), the first cover (**221**) being electrically insulated from the at least one first current collector (**245**); [0127] a second plurality of cylindrical battery cells (**230**) disposed in a second chamber (**229**) defined by the housing laterally between the cooling channel (**226**) and the second cover (**223**), each battery cell of the second plurality of battery cells (**230**) extending generally orthogonally to the cooling channel (**226**) and the second cover (**223**); and [0128] at least one second current collector (**245**) disposed laterally between the second cover (**223**) and the second plurality of battery cells (**230**), the at least one second current collector (**245**) being electrically connected to the second plurality of battery cells (**230**), [0129] the second cover (**223**) enclosing the at least one second current collector (**245**) and outer ends of the battery cells (**230**) of the second plurality of battery cells (**230**), the second cover (**223**) being electrically insulated from the at least one second current collector (**245**).

[0130] CLAUSE 2. The vehicle (**100**, **300**) of clause 1, wherein: [0131] the battery cells (**230**) of the first plurality of battery cells (**230**) are separated into a first plurality of battery modules (**235**); [0132] the at least one first current collector (**245**) includes a first plurality of current collectors (**245**) connected in series; [0133] each one of the first plurality of current collectors (**245**) is electrically connected to a corresponding battery module (**235**) of the first plurality of battery modules (**235**); [0134] the battery cells (**230**) of the second plurality of battery cells (**230**) is separated into a second plurality of battery modules (**235**); [0135] the at least one second current collector (**245**) includes a second plurality of current collectors (**245**) connected in series; and [0136] each one of the second plurality of current collectors (**245**) is electrically connected to a corresponding battery module (**235**) of the second plurality of battery modules (**235**).
[0137] CLAUSE 3. The vehicle (**100**, **300**) of clause 2, wherein the second plurality of battery modules (**235**) has a greater number of battery modules (**235**) than the first plurality of battery modules (**235**).
[0138] CLAUSE 4. The vehicle (**100**, **300**) of clause 3, wherein: [0139] the first plurality of battery modules (**235**) includes three first battery modules (**235**) disposed in the first chamber (**225**); [0140] the first plurality of current collectors (**245**) includes three first current collectors (**245**); [0141] the second plurality of battery modules (**235**) includes four second battery modules (**235**) disposed in the second chamber (**229**); and [0142] the second plurality of current collectors (**245**) includes four second current collectors (**245**).
[0143] CLAUSE 5. The vehicle (**100**, **300**) of clause 4, wherein: [0144] the three first current collectors (**245**) are electrically connected to one another in series; and [0145] the four second current collectors (**245**) are electrically connected to one another in series.
[0146] CLAUSE 6. The vehicle (**100**, **300**) of clause 2, wherein: [0147] the first plurality of current collectors (**245**) is a first plurality of printed circuit boards (PCBs) (**245**); and [0148] the second plurality of current collectors (**245**) is a second plurality of PCBs (**245**).
[0149] CLAUSE 7. The vehicle (**100**, **300**) of clause 6, wherein each PCB (**245**) of the first and second pluralities of PCBs (**245**) includes a module battery management system (BMS) (**243**) for managing the corresponding battery module (**235**) of the first and second pluralities of battery modules (**235**).
[0150] CLAUSE 8. The vehicle (**100**, **300**) of clause 3, wherein the battery pack (**210**) further comprises at least one of: [0151] a global BMS (**240**) disposed in the first chamber (**225**), the global BMS (**240**) being electrically connected to the at least one first current collector (**245**) and the at least one second current collector (**245**); and [0152] a DC-DC converter (**242**) disposed in the first chamber (**225**), the DC-DC converter (**242**) being electrically connected to at least one of the at least one first current collector (**245**) and the at least one second current collector (**245**).
[0153] CLAUSE 9. The vehicle (**100**, **300**) of any one of clauses 1 to 8, wherein: [0154] the electric powerpack (**200**) further comprises a plurality of electrical components; [0155] at least one of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells (**230**) and disposed on an exterior of the battery housing (**220**); and [0156] at least one other of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells (**230**) and disposed on an interior of the battery housing (**220**).
[0157] CLAUSE 10. The vehicle (**100**, **300**) of any one of clauses 1 to 9, wherein the electric powerpack (**200**) further comprises at least one of: [0158] a charger (**250**) electrically connected to the first and second pluralities of cylindrical battery cells (**230**); and [0159] an inverter (**260**) electrically connected to the first and second pluralities of cylindrical battery cells (**230**).
[0160] CLAUSE 11. The vehicle (**100**, **300**) of clause 10, wherein the electric powerpack (**200**) comprises both the charger (**250**) and the inverter (**260**).
[0161] CLAUSE 12. The vehicle (**100**, **300**) of clause 11, wherein the charger (**250**) and the inverter (**260**) are mounted to the battery housing (**220**).

[0162] CLAUSE 13. The vehicle (**100**, **300**) of any one of clauses 1 to 12, wherein: [0163] the housing body (**227**) includes: [0164] a first lateral portion (**227**A), and [0165] a second lateral portion (**227**B) connected to the first lateral portion (**227**A); [0166] the first cover (**221**) being selectively connected to the first lateral portion (**227**A); [0167] the second cover (**223**) being selectively connected to the second lateral portion (**227**B); [0168] an inner face of the first lateral portion (**227**A) including a first channel form (**226**A); [0169] an inner face of the second lateral portion (**227**B) including a second channel form (**226**B); [0170] the cooling channel (**226**) is defined by a space created between the inner face of the first lateral portion (**227**A) and the inner face of the second lateral portion (**227**B); and [0171] the center portion of the housing body (**227**) being formed by an inner part of the first lateral portion (**227**A) and an inner part of the second lateral portion (**227**B).
[0172] CLAUSE 14. The vehicle (**100**, **300**) of any one of clauses 1 to 13, wherein the battery pack (**210**) further comprises: [0173] at least one first dielectric layer (**248**) disposed between inner ends of the first plurality of battery cells (**230**) and the center portion the housing body (**227**); and [0174] at least one second dielectric layer (**248**) disposed between inner ends of the second plurality of battery cells (**230**) and the center portion the housing body (**227**).
[0175] CLAUSE 15. The vehicle (**100**, **300**) of any one of clauses 1 to 14, wherein, when in operation, coolant flowing through the cooling channel (**226**) absorbs heat from inner ends of the first plurality of battery cells (**230**) and inner ends of the second plurality of battery cells (**230**).
[0176] CLAUSE 16. The vehicle (**100**, **300**) of any one of clauses 1 to 15, wherein: [0177] inner ends of the first plurality of battery cells (**230**) are in thermal communication with the cooling channel (**226**) through the housing body (**227**); and [0178] inner ends of the second plurality of battery cells (**230**) are in thermal communication with the cooling channel (**226**) through the housing body (**227**).
[0179] CLAUSE 17. The vehicle (**100**, **300**) of any one of clauses 1 to 16, further comprising a liquid cooling system including: [0180] a fluid pump (**278**); [0181] a coolant reservoir (**270**); and [0182] a cooling circuit (**290**) formed at least in part by the cooling channel (**226**) of the battery housing (**220**), the cooling channel (**226**) being fluidly connected to the fluid pump (**278**) and the coolant reservoir (**270**).
[0183] CLAUSE 18. The vehicle (**100**, **300**) of any one of clauses 1 to 17, wherein: [0184] the at least one ground-engaging member (**121**, **127**) includes: [0185] a front wheel (**121**) disposed at least in part forward of the powerpack (**200**), and [0186] a rear wheel (**127**) disposed at least in part rearward of the powerpack (**200**); and [0187] the vehicle (**100**) is an electric motorcycle (**100**).
[0188] CLAUSE 19. The vehicle (**100**, **300**) of any one of clauses 1 to 18, wherein: [0189] the vehicle (**100**, **300**) is an electric snowmobile (**300**); and [0190] the at least one ground-engaging member includes two skis (**326**).
[0191] CLAUSE 20. A straddle seat electric vehicle (**100**, **300**) comprising: [0192] a frame (**110**); [0193] a straddle seat (**140**) supported by the frame (**110**); [0194] at least one ground-engaging member (**121**, **127**) operatively connected to the frame (**110**); [0195] an electric motor (**160**) being operatively connected to the at least one ground-engaging member (**121**, **127**) for driving the at least one ground-engaging member (**121**, **127**); [0196] an electric powerpack (**200**) operatively connected to the electric motor (**160**), the powerpack (**200**) comprising: [0197] a battery pack (**210**) comprising: [0198] a battery housing (**220**) including: [0199] a housing body (**227**), [0200] a cooling channel (**226**) defined in the housing body (**227**), the cooling channel (**226**) extending generally vertically through a center portion of the housing body (**227**), [0201] a first cover (**221**) selectively connected to the housing body (**227**), and [0202] a second cover (**223**) selectively connected to the housing body (**227**); [0203] a first plurality of battery modules (**235**) disposed in a first chamber (**225**) defined by the housing laterally between the cooling channel (**226**) and the first cover (**221**), each module (**235**) of the first plurality of battery modules (**235**) including a plurality of battery cells (**230**); [0204] a plurality of first current collectors (**245**) disposed laterally between

the first cover (**221**) and the first plurality of battery modules (**235**), each battery module (**235**) of the first plurality of battery modules (**235**) being operatively connected to a corresponding one of the first plurality of current collectors (**245**), [0205] the plurality of battery cells (**230**) of each battery module (**235**) of the first plurality of battery modules (**235**) being connected together in series by the corresponding one of the first plurality of current collectors (**245**); [0206] a second plurality of battery modules (**235**) disposed in a second chamber (**229**) defined by the housing laterally between the cooling channel (**226**) and the second cover (**223**), each module (**235**) of the second plurality of battery modules (**235**) including a plurality of battery cells (**230**); and [0207] a plurality of second current collectors (**245**) disposed laterally between the second cover (**223**) and the second plurality of battery modules (**235**), each battery module (**235**) of the second plurality of battery modules (**235**) being operatively connected to a corresponding one of the second plurality of current collectors (**245**), [0208] the plurality of battery cells (**230**) of each battery module (**235**) of the second plurality of battery modules (**235**) being connected together in series by the corresponding one of the second plurality of current collectors (**245**).
[0209] CLAUSE 21: A battery pack for an electric vehicle, the battery pack comprising: [0210] a battery housing; [0211] a general battery management system (general BMS) disposed in the battery housing; and [0212] a plurality of battery modules disposed in the battery housing, each battery module of the plurality of battery modules comprising: [0213] a plurality of battery cells; and a module board comprising: [0214] a module battery management system (module BMS) communicatively connected to the general BMS, the module BMS including at least one sensor for sensing at least one operating condition of the plurality of battery cells; and
an integrated current collector disposed within the battery module, the integrated current collector electrically coupling the plurality of battery cells together.
[0215] Modifications and improvements to the above-described implementations of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A straddle seat electric vehicle comprising: a frame; a straddle seat supported by the frame; at least one ground-engaging member operatively connected to the frame; an electric motor being operatively connected to the at least one ground-engaging member for driving the at least one ground-engaging member; an electric powerpack operatively connected to the electric motor, the powerpack comprising: a battery pack comprising: a battery housing including: a housing body, a cooling channel defined in the housing body, the cooling channel extending generally vertically through a center portion of the housing body, a first cover selectively connected to the housing body, and a second cover selectively connected to the housing body; a first plurality of cylindrical battery cells disposed in a first chamber defined by the battery housing laterally between the cooling channel and the first cover, each battery cell of the first plurality of battery cells extending generally orthogonally to the cooling channel and the first cover; at least one first current collector disposed laterally between the first cover and the first plurality of battery cells, the at least one first current collector being electrically connected to the first plurality of battery cells, the first cover enclosing the at least one first current collector and outer ends of the battery cells of the first plurality of battery cells, the first cover being electrically insulated from the at least one first current collector; a second plurality of cylindrical battery cells disposed in a second chamber defined by the battery housing laterally between the cooling channel and the second cover, each battery cell of the second plurality of battery cells extending generally orthogonally to the cooling channel and the second cover; and at least one second current collector disposed laterally between the second cover and the second plurality of battery cells, the at least one second current collector being electrically connected to the second plurality of battery cells, the second cover enclosing the at least one second current collector and outer ends of the battery cells of the second plurality of battery cells, the second cover being electrically insulated from the at least one second current collector.

2. The vehicle of claim 1, wherein: the battery cells of the first plurality of battery cells are separated into a first plurality of battery modules; the at least one first current collector includes a first plurality of current collectors connected in series; each one of the first plurality of current collectors is electrically connected to a corresponding battery module of the first plurality of battery modules; the battery cells of the second plurality of battery cells is separated into a second plurality of battery modules; the at least one second current collector includes a second plurality of current collectors connected in series; and each one of the second plurality of current collectors is electrically connected to a corresponding battery module of the second plurality of battery modules.

3. The vehicle of claim 2, wherein the second plurality of battery modules has a greater number of battery modules than the first plurality of battery modules.

4. The vehicle of claim 3, wherein: the first plurality of battery modules includes three first battery modules disposed in the first chamber; the first plurality of current collectors includes three first current collectors; the second plurality of battery modules includes four second battery modules disposed in the second chamber; and the second plurality of current collectors includes four second current collectors.

5. The vehicle of claim 4, wherein: the three first current collectors are electrically connected to one another in series; and the four second current collectors are electrically connected to one another in series.

6. The vehicle of claim 2, wherein: the first plurality of current collectors is a first plurality of printed circuit boards (PCBs); and the second plurality of current collectors is a second plurality of PCBs.

7. The vehicle of claim 6, wherein each PCB of the first and second pluralities of PCBs includes a module battery management system (BMS) for managing the corresponding battery module of the first and second pluralities of battery modules.

8. The vehicle of claim 3, wherein the battery pack further comprises at least one of: a global BMS disposed in the first chamber, the global BMS being electrically connected to the at least one first current collector and the at least one second current collector; and a DC-DC converter disposed in the first chamber, the DC-DC converter being electrically connected to at least one of the at least one first current collector and the at least one second current collector.

9. The vehicle of claim 1, wherein: the electric powerpack further comprises a plurality of electrical components; at least one of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells and disposed on an exterior of the battery housing; and at least one other of the plurality of electrical components being electrically connected to the first and second pluralities of cylindrical battery cells and disposed on an interior of the battery housing.

10. The vehicle of claim 1, wherein the electric powerpack further comprises at least one of: a charger electrically connected to the first and second pluralities of cylindrical battery cells; and an inverter electrically connected to the first and second pluralities of cylindrical battery cells.

11. The vehicle of claim 10, wherein the electric powerpack comprises both the charger and the inverter.

12. The vehicle of claim 11, wherein the charger and the inverter are mounted to the battery housing.

13. The vehicle of claim 1, wherein: the housing body includes: a first lateral portion, and a second lateral portion connected to the first lateral portion; the first cover being selectively connected to the first lateral portion; the second cover being selectively connected to the second lateral portion; an inner face of the first lateral portion including a first channel form; an inner face of the second lateral portion including a second channel form; the cooling channel is defined by a space created between the inner face of the first lateral portion and the inner face of the second lateral portion; and the center portion of the housing body being formed by an inner part of the first lateral portion and an inner part of the second lateral portion.

14. The vehicle of claim 1, wherein the battery pack further comprises: at least one first dielectric layer disposed between inner ends of the first plurality of battery cells and the center portion the housing body; and at least one second dielectric layer disposed between inner ends of the second plurality of battery cells and the center portion the housing body.

15. The vehicle of claim 1, wherein, when in operation, coolant flowing through the cooling channel absorbs heat from inner ends of the first plurality of battery cells and inner ends of the second plurality of battery cells.

16. The vehicle of claim 1, wherein: inner ends of the first plurality of battery cells are in thermal communication with the cooling channel through the housing body; and inner ends of the second plurality of battery cells are in thermal communication with the cooling channel through the housing body.

17. The vehicle of claim 1, further comprising a liquid cooling system including: a fluid pump; a coolant reservoir; and a cooling circuit formed at least in part by the cooling channel of the battery housing, the cooling channel being fluidly connected to the fluid pump and the coolant reservoir.

18. The vehicle of claim 1, wherein: the at least one ground-engaging member includes: a front wheel disposed at least in part forward of the powerpack, and a rear wheel disposed at least in part rearward of the powerpack; and the vehicle is an electric motorcycle.

19. The vehicle of claim 1, wherein: the vehicle is an electric snowmobile; and the at least one ground-engaging member includes two skis.

20. A straddle seat electric vehicle comprising: a frame; a straddle seat supported by the frame; at least one ground-engaging member operatively connected to the frame; an electric motor being operatively connected to the at least one ground-engaging member for driving the at least one ground-engaging member; an electric powerpack operatively connected to the electric motor, the powerpack comprising: a battery pack comprising: a battery housing including: a housing body, a cooling channel defined in the housing body, the cooling channel extending generally vertically through a center portion of the housing body, a first cover selectively connected to the housing body, and a second cover selectively connected to the housing body; a first plurality of battery modules disposed in a first chamber defined by the housing laterally between the cooling channel and the first cover, each module of the first plurality of battery modules including a plurality of battery cells; a plurality of first current collectors disposed laterally between the first cover and the first plurality of battery modules, each battery module of the first plurality of battery modules being operatively connected to a corresponding one of the first plurality of current collectors, the plurality of battery cells of each battery module of the first plurality of battery modules being connected together in series by the corresponding one of the first plurality of current collectors; a second plurality of battery modules disposed in a second chamber defined by the housing laterally between the cooling channel and the second cover, each module of the second plurality of battery modules including a plurality of battery cells; and a plurality of second current collectors disposed laterally between the second cover and the second plurality of battery modules, each battery module of the second plurality of battery modules being operatively connected to a corresponding one of the second plurality of current collectors, the plurality of battery cells of each battery module of the second plurality of battery modules being connected together in series by the corresponding one of the second plurality of current collectors.

21. (canceled)